

Measurement of Electricity

Electrical Pressure

- Volts are the measurements of units for voltage
- Voltage is the energy between points in a circuit (how much it flows)
- Resistance is how much they are opposed to the flow
- The higher the voltage the more electricity will flow through the circuit
- If this is too high or too low it can cause a circuit to damage

Voltage is pressure and is measured in volts

Electrical Power

- Wattage is the amount of power used for a device to work
- Wattage are measured in Watts (the units of power)

Electrical Current Strength

- The measure of current during a specific time (ie electricity in a circuit in 2 seconds)
- Amps/amperes is unit of electric current. 1 ampere = 1 coulomb
- Milli- amps = 1000 of an amp
- Amperage – force of current measured in amps

Recap the Principles of Electric Currents and Basic Electricity

- The flow of an electric charge is referred to as **electric current**.
- Current is measured in amperes and there are two different types:
 - Direct Current (DC) – current flows in one direction with a constant voltage polarity
 - Alternating Current (AC) – current that changes direction periodically along with its voltage polarity
- The unit of electrical **power** is **watts**.
- The unit of electrical **strength** is **volts**.
- The unit of electrical **resistance** is an **ohm**.

The Principles of Electric Currents and Basic Electricity

- One of the fundamental laws of electricity is Ohms law. This states that the voltage through a conductor between two points is directly proportional to the current flowing through it.
- How to identify the correct fuse used with different electrical appliances:
 - 3 amp – appliances up to 700 watts
 - 5 amp – appliances between 750-1000 watts
 - 13 amp – appliances between 1000-1300 watts
- To identify the correct fuse, manufacturers work from the equation of:
- **Power (watts) = Voltage x Current (Amps)**

Ohm

- Unit of electrical resistance
- Named after the 19th century German physicist Georg Ohm

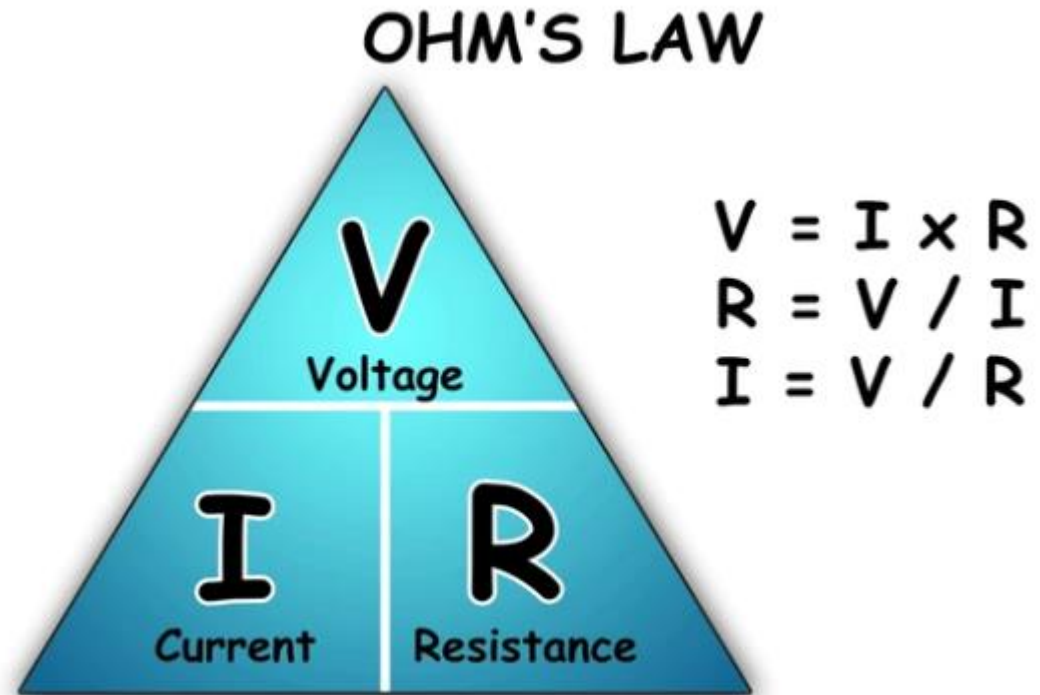
I is the current through the conductor

V is the voltage measured

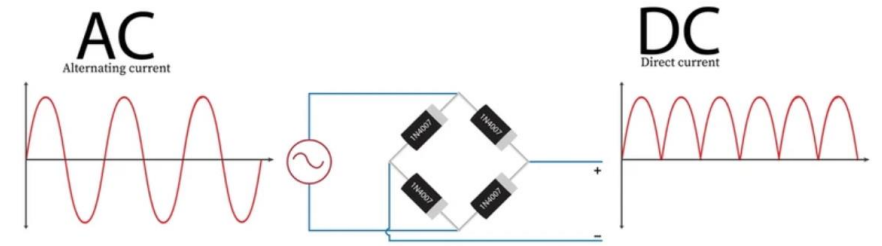
R is the resistance

Ohm's Law

A method used to calculate the relationship between voltage, current and resistance in electrical circuits



Rectifiers



A rectifier is an electrical device that **converts alternating current (AC) to direct current (DC)**. Rectifiers are used in many electronic devices, power supplies, and electrical systems to transform AC, which is commonly available from power outlets, into DC, which is needed by most electronics and devices.

There are two main types of rectifiers:

1. Half-Wave Rectifier:

1. This type of rectifier only allows one half of the AC signal (either the positive or negative half) to pass through, blocking the other half.
2. It uses a single diode to rectify the signal.
3. The output is not pure DC; it has a pulsating DC waveform.

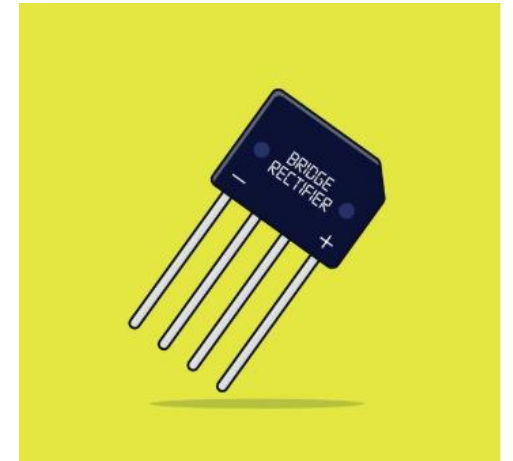
2. Full-Wave Rectifier:

1. A full-wave rectifier allows both the positive and negative halves of the AC signal to be used.
2. This is achieved using two diodes in a **center-tapped transformer** configuration or four diodes arranged in a **bridge rectifier**.
3. The output of a full-wave rectifier is smoother and closer to a pure DC compared to a half-wave rectifier.

Rectifier Circuits

Rectifier Circuits:

- **Bridge Rectifier:** A commonly used arrangement for full-wave rectification. It uses four diodes arranged in a bridge to connect both halves of the AC signal.
- **Center-Tapped Full-Wave Rectifier:** This setup uses a transformer with a center-tapped secondary coil and two diodes, with the center tap providing a reference point.



Capacitors

- A **capacitor** is an electronic component that stores and releases electrical energy in the form of an electric field.
- It consists of two conductive plates separated by an insulating material known as a **dielectric**.
- When a voltage is applied across the plates, the capacitor stores electrical energy. When the voltage is removed, the capacitor can release this stored energy.

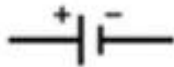
Electrical Circuit Symbols



Switch (closed)



Switch (open)



Cell



Resistor



Capacitor



Inductor



Buzzer



Wire



Resistor



Battery



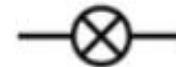
Fuse



Variable Resistor



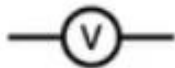
Thermistor



Lamp



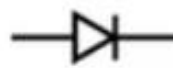
Ammeter



Voltmeter



Motor



Diode



LDR



Ground

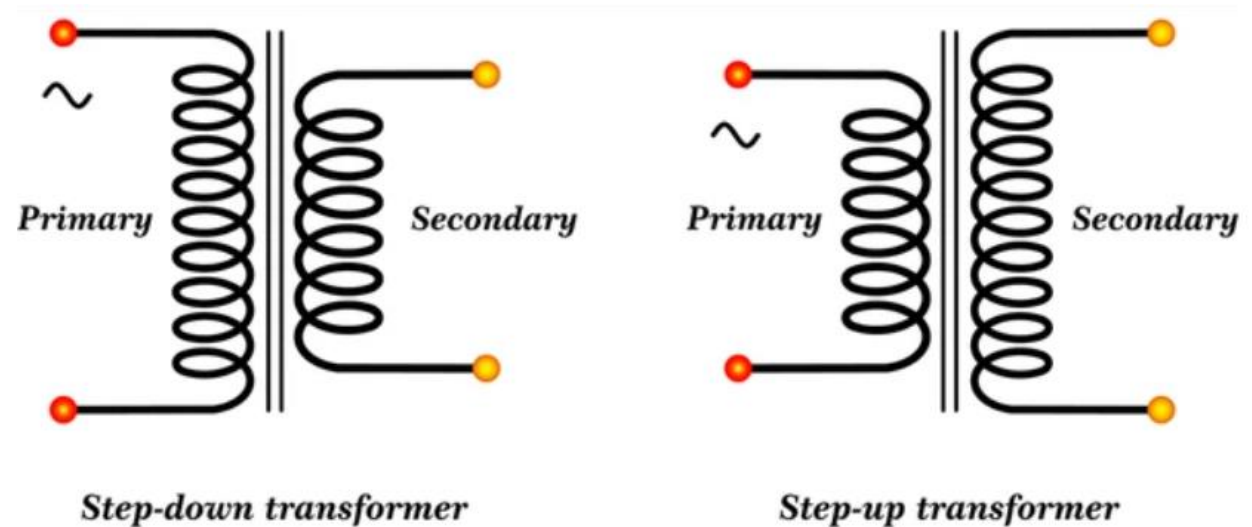


Transistor

Transformers

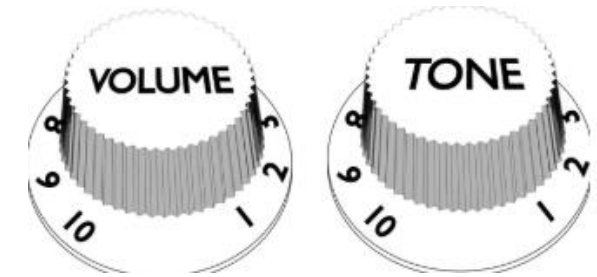
A **transformer** is an electrical device used to change the voltage of an alternating current (AC) signal.

It works on the principle of electromagnetic induction and is widely used in power distribution systems to step up (increase) ↑ or step down (decrease) ↓ voltage levels efficiently.



Potentiometer

A type of variable resistor used to control the voltage or current in a circuit by adjusting its resistance. It's commonly referred to as a "pot" and is used in many electronic applications, such as adjusting the volume on audio equipment, controlling the brightness of lights, or in circuits requiring slight adjustments of resistance.



A potentiometer consists of three main components:

1. Resistive Track (or Element): This is a length of resistive material (usually carbon or a metal oxide) that has a specific resistance value. Wiper: A movable contact that slides along the resistive track. The position of the wiper determines the resistance between the wiper and the two fixed terminals.
2. Terminals:
 1. Two fixed terminals are connected to the ends of the resistive track.
 2. One movable terminal is connected to the wiper and provides the variable resistance output.

Rheostat

- A **rheostat** is a type of variable resistor used to control the current in a circuit by adjusting its resistance.
- It is similar to a potentiometer with only two terminals and is primarily used to vary the flow of current in a circuit, rather than controlling the voltage.



Starter

- An electrical starter is a device used to safely start and control the operation of an electric motor.
- It regulates the current flow,
- Protects the motor from overloads, (sometimes also helps reduce the voltage to prevent damage
- Connects the motor to the power supply.
- Stops the Motor and disconnects the motor safely.
- Reduces Start-Up Current (limits inrush current for some types) especially in large motors

Oudin Coil – Resonator

- An Oudin coil is a type of high-voltage, high-frequency resonant transformer, closely related to the Tesla coil.
- It was invented by French physician Paul Marie Oudin in the late 19th century and is primarily used for medical electrotherapy (historically)
- It produces high-frequency alternating current (AC). The high voltage can reach tens to hundreds of kilovolts.
- There is a single electrode and it typically has a single output terminal.
- It uses resonant inductive coupling between a primary coil and a secondary coil. A spark gap or oscillator circuit rapidly switches the current in the primary coil. This induces a resonant voltage in the secondary coil, producing high-frequency, high-voltage AC at the output.
- Applications (Historical and Modern)Electrotherapy (early 1900s): Used in diathermy and "violet ray" treatments.

Insulators

- An insulator is a material that resists the flow of electric current.
- It prevents electricity from escaping its intended path, making it essential for safety, reliability, and efficiency in electrical systems.

Examples of a good insulator;

High electrical resistance

Low thermal conductivity

Durability and heat resistance

Non-conductive atoms (tight electron bonding)



Electromagnetic Radiation

- Electromagnetic (EM) radiation is a type of energy that travels through space in the form of waves. It includes everything from visible light to radio waves, microwaves, X-rays, and more.
- It moves in two ways at the same time—like an electric wave and a magnetic wave, which travel together at the speed of light.
- EM radiation doesn't need air, water, or any other medium to travel. That's why sunlight reaches Earth through space.

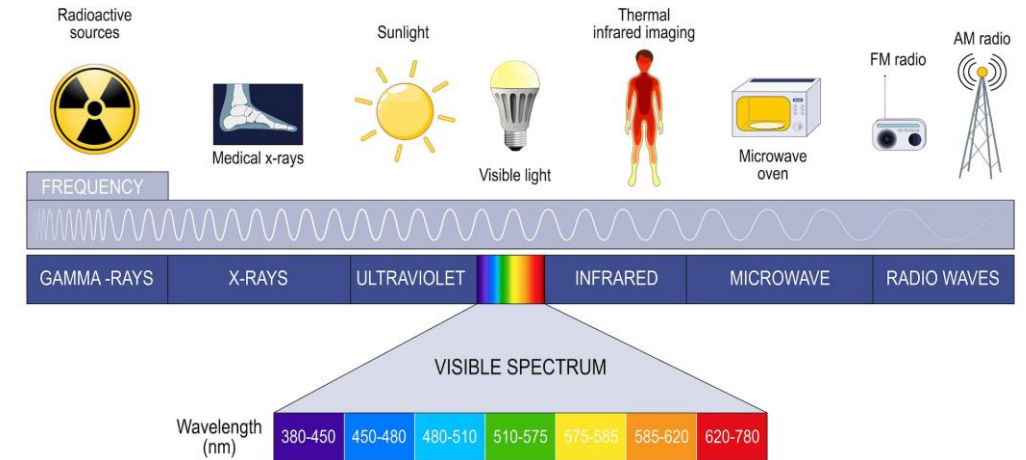
EM Radiation

- 3. Types of EM Radiation:
 - The different types are arranged in what's called the electromagnetic spectrum, from longest to shortest wavelength:
 - Radio waves – Used for communication (like radios, TVs, and WiFi).
 - Microwaves – Used for cooking and mobile phones.
 - Infrared – Felt as heat (like from the sun or a heater).
 - Visible light – The only part we can see (rainbows, light bulbs).
 - Ultraviolet (UV) – Causes sunburn and is used for sterilization.
 - X-rays – Used in medical imaging to see bones.
 - Gamma rays – Very powerful and used in cancer treatment.

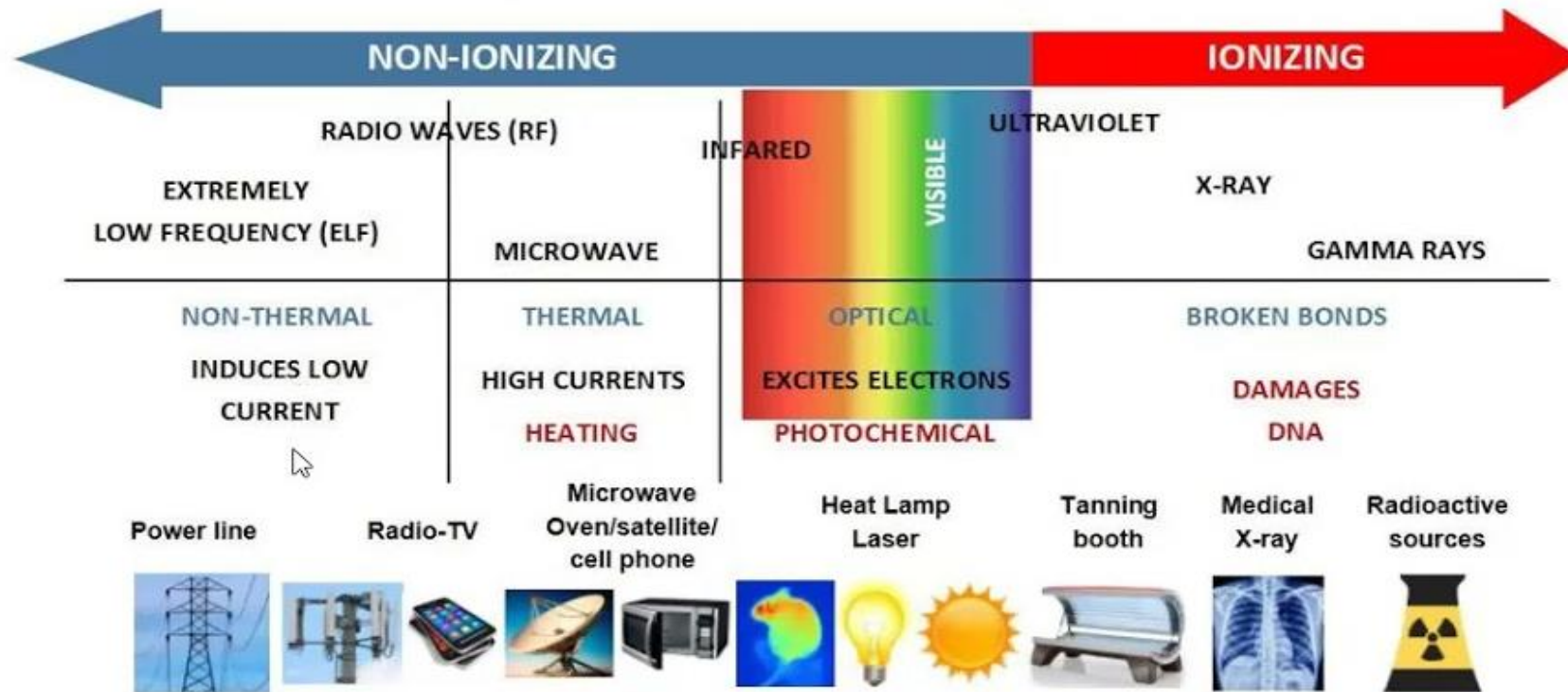
Electromagnetic Spectrum

- The **electromagnetic (EM) spectrum** is the range of all types of electromagnetic radiation.
- Electromagnetic radiation consists of waves of electric and magnetic fields that oscillate in space and travel at the speed of light.
- These waves vary in wavelength and frequency, and the electromagnetic spectrum is a way of organizing them based on these properties.

Electromagnetic spectrum

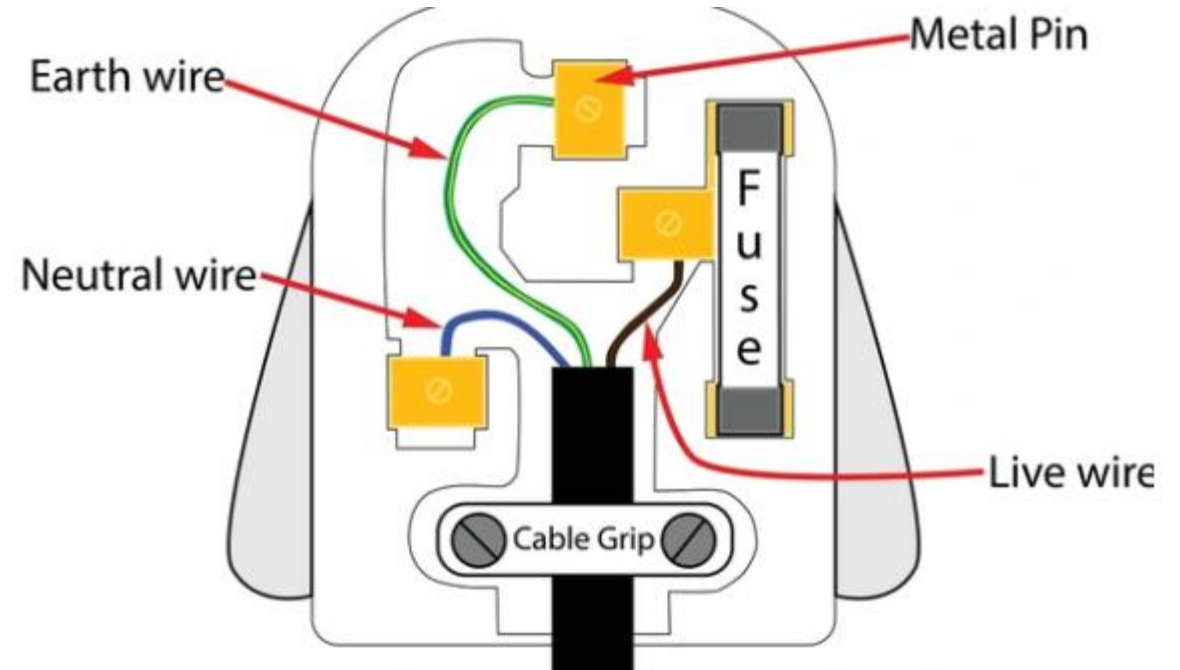


Non-ionising & Ionising Radiation



Main's Plug

- Live = Brown
- Neutral = Blue
- Earth = Yellow & Green



Portable Appliance Testing (PAT)

- A process which ensures the safety of electrical appliances
- The process is usually required annually and involves both visual inspections and electrical testing to identify any safety issues or possible hazards.
- Best practice is to ensure your equipment meets safety standards